

PHAROS AAI 2027 Technical Supplement

Paper Title: PHAROS: Navigated Hierarchical Attention with Adaptive Budgets, Cross-Level Guidance, Tidal Momentum, and Uncertainty-Aware Fallback

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Supplement Scope

This technical supplement provides additional derivations and implementation details supporting the PHAROS main paper. The main paper is self-contained; this supplement is intended only as optional supporting material for reviewers who want additional proof intuition, complexity details, and evaluation-planning information.

A. Proof Details for Depth-Sounding Budget Allocation

Depth-Sounding Budget Allocation (DBA) redistributes a fixed global token-selection budget across attention heads according to selection entropy. Let the global selection budget be k_{total} and let the normalized per-head allocation weights be w_h , where $\sum_h w_h = 1$. Ignoring rounding and minimum-budget correction, DBA assigns $k_h = k_{\text{total}} * w_h$. Therefore, $\sum_h k_h = \sum_h k_{\text{total}} * w_h = k_{\text{total}} * \sum_h w_h = k_{\text{total}}$. Thus, DBA is cost-neutral under exact allocation: it changes the distribution of selected tokens across heads but does not change the global token-selection budget.

A.1 Recall Intuition

Let $R_h(k_h)$ denote expected recall for head h under budget k_h . Focused low-entropy heads tend to saturate quickly because only a small number of candidate tokens dominate selection probability. Diffuse high-entropy heads tend to benefit from additional selected tokens because attention mass is spread across more candidates. If R_h is concave in k_h , transferring budget from saturated low-entropy heads to unsaturated high-entropy heads does not reduce total expected recall and may improve it when entropy heterogeneity is present.

B. Tidal Momentum Selection Stability

Tidal Momentum Selection (TMS) introduces an exponential moving average over selection logits across transformer depth. If the variance of selection logits is bounded and the momentum coefficient is stable, the EMA reduces high-frequency layer-to-layer fluctuations because each tidal state combines current evidence with previous-layer evidence. This supports more stable token selection without introducing a full recurrent hidden state or changing the attention kernel.

C. Fog Gate Confidence Fallback

Fog Gate Confidence Fallback (FCF) expands the selected token beam when selector confidence is low. Let confidence be $c_{\{h,t\}} = \max_j \text{softmax}(s_{\{h,t\}})_j$. If confidence falls below a learned threshold, PHAROS expands the local budget from k_h to $\min(2k_h, P)$. The expected selected-token count is $E[k_{\text{fog}}] = (1 - f) * k_h + f * \min(2k_h, P)$, where f is the empirical fallback activation rate. If f is small, expected cost remains close to the baseline budget.

D. Complexity Derivation

PHAROS is designed as a thin wrapper over Lighthouse-style hierarchical selection. Its additional operations are: RCG coarse-to-fine upsampling and additive logit bias; TMS exponential moving average update over selection logits; DBA entropy computation and periodic budget update; and FCF confidence computation and fallback-triggered budget expansion. These operations do not change the asymptotic structure of the hierarchical selection pipeline.

E. Recommended Evaluation Details

A complete empirical evaluation should compare PHAROS against dense SDPA, Lighthouse Attention with norm scorer, Lighthouse Attention with dilated scorer, PHAROS without each mechanism, and full PHAROS. Recommended metrics include training loss, resume loss, validation perplexity, long-context retrieval accuracy, throughput, memory usage, wall-clock latency, fallback activation rate, and per-head selected-token distribution.

F. Reporting Policy

The current PHAROS submission does not report verified quantitative results. This supplement therefore does not include fabricated or placeholder metrics. Any future empirical version should report model size, tokenizer, dataset mixture, context length, training tokens, optimizer settings, seeds, GPU hardware, runtime, variance across runs, and statistical confidence where appropriate.